Adi Nata

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LinkedIn | GitHub | Portfolio



PERSONAL SUMMARY

Software engineer with experience building full-stack web applications, distributed backend services, and secure cloud platforms. Hands-on experience with Node.js, Python, React, PostgreSQL, and AWS across manufacturing, research and government environments supported by Agile development experience and AWS and Kubernetes certifications. Delivered MES applications, automated machine workflows, and a secure data-sharing platform serving more than 1500 users.

SKILLS

- Programming Languages: TypeScript, JavaScript, Python, C, C++, C#.
- Web development: React, Svelte, Node.js, HTML, Tailwind, Spring Boot, .NET.
- Database: PostgreSQL, MySQL, MongoDB, Redis.
- DevOps: Git, Docker, Kubernetes, Terraform, AWS, OCI, GitHub Actions.
- Core Competencies: Problem Solving, Fast Learner, Communication, Collaboration, Adaptability, Mobility.

EXPERIENCES

Senior Research Engineer

*A*Star SIMTech, Jan 2026 - Present*

- Owned end-to-end delivery of a secure data-sharing platform connecting more than 20 institutes with more than 1500 users, from requirements gathering and architecture through implementation, testing, and deployment. Built the React frontend and MCP-compliant Node.js backend, provisioned AWS infrastructure with Terraform, and automated deployment through GitHub Actions.
- Designed and implemented a distributed task orchestration architecture composed of microservices, asynchronous messaging, and API-driven workflow coordination to automate execution across heterogeneous machine systems with .NET, Python, RabbitMQ and MySQL logging database.
- Developed SiLA compliant (gRPC-based) hardware adapter library to support standardized machine communication from underlying machine protocol (TCP/IP, serial communication, MODBUS).
- Deployed and maintained a containerized data pipeline using Kafka and Debezium CDC to capture, retain, and process manufacturing data from 300 sensors, with 500 ms end-to-end latency from machine data output to ingestion into the TimescaleDB data lake.
- Developed a laboratory equipment data pipeline with OpenTelemetry monitoring to process real-time telemetry, retain historical data, and track equipment health and pipeline performance.

Software Engineer

Vital Thin Film Materials, Apr 2024 – Dec 2025

- Owned full-stack MES development across React dashboards, Node.js backend services, and PostgreSQL databases, continuously streaming machine data and execution status to the frontend for real-time monitoring.
- Developed a C++ CNC data ingestion and backend service to process and store machine data in TimescaleDB and central NFS storage, alongside a monitoring

service that detected OPC UA task-completion events and notified the Node.js backend via MQTT.
- Configured network infrastructure, including Meraki router and firewall setup, wireless heatmap creation, and VPN connection planning.
- Authored IT policies such as Disaster Recovery Plan, Incident Response Plan, and Cyber Security Awareness, ensuring compliance with ISO 27701.

Embedded Software Engineer
Continental Automotive, Feb 2023 - Mar 2024
- Developed a high-performance C and C++ network communication library to enable secure HTTP and MQTT connections between vehicle firmware and central servers, following Agile methodology.
- Enhanced and automated the firmware build process by optimizing the CMake command and creating bash scripts, reducing compilation and build time from 15 minutes to 5 minutes.

EDUCATION

National University of Singapore (NUS), Computer Engineering
- Minor: Financial Mathematics.
- Activities: Engineering Student Ambassadors.
- CAP: 4.17/5 (Second Class Honors).

CERTIFICATIONS

- Certified Kubernetes Administrator (CKA)
- AWS Certified Solutions Architect - Professional
- Claude Certified Architect - Foundations
- OCI Certified Developer - Professional
- Certified SAFe 6 Practitioner

ADDITIONAL EXPERIENCES

Research Assistant
National University of Singapore, Jan 2022 - Jun 2022
- Developed a CNN model to detect malaria-infected cells with 91% accuracy and 95% precision.

Business Analyst Intern
Workforce Optimizer Pte Ltd, Jun 2021 - Dec 2021
- Enhanced LSTM forecasting model for manpower allocation, reducing RMSE from 3.4 to 1.2.

Teaching Assistant
National University of Singapore, Jun 2020 – Jun 2022
- Mentored over 100 students in C programming with average student rating of 4.7/5.

LANGUAGES

- English (Bilingual)
- Indonesian (Native)
- Malay (Native)
- Mandarin (Professional)